A Theory of Continuous Attractor Manifold Learning for Spatial Cognition

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Abstract

Spatial cognition requires neural representations that remain stable under perturbations while tracking changes in orientation and location. Continuous attractor manifolds provide this dynamical structure, but learning dynamics that stabilize and move along them remains challenging. We propose Continuous Attractor Manifold Learning (CAML), an energy-based framework for learning these dynamics from population codes. CAML shapes a scalar energy whose low-energy set represents a supplied spatial manifold, with its gradient restoring perturbed activity and a separately learned bump-shifting field driving motion along the manifold. We establish autonomous energy descent and, for sufficiently expressive energy families, uniform approximation of smooth target energies and their gradients. Experiments on head-direction rings, grid-cell tori, and a bounded non-periodic place manifold demonstrate perturbation recovery and velocity-driven tracking. Local excitation and surround inhibition also emerge from the learned restoring dynamics. CAML thus links continuous attractor theory to trainable dynamics for spatial representation.

1 Introduction

Spatial cognition depends on neural representations of orientation and location that remain stable while tracking self-motion and sensory cues [1, 2, 3, 4]. Population recordings have revealed low-dimensional manifolds in these systems, including a ring-like manifold in the head-direction circuit and a toroidal manifold in grid-cell modules [5, 6]. These findings motivate continuous attractors as a principle for spatial representation, where recurrent dynamics restore perturbations toward a manifold while allowing input-driven motion along it [7, 8, 2, 6].

Classical continuous attractor neural networks (CANNs) realize this principle by combining recurrent interactions that stabilize a continuum of activity states with input-dependent components that translate activity along the attractor [9, 10, 11]. This logic has been used to model head-direction rings and grid-cell path integration through structured recurrent architectures whose population states are shifted by angular or linear velocity inputs [7, 12, 13, 8, 14]. Despite their mechanistic clarity, these models depend on carefully specified recurrent kernels, symmetry assumptions, boundary conditions, and drive terms, which determine whether the attractor exists, remains stable, and accurately tracks inputs [8, 11, 10].

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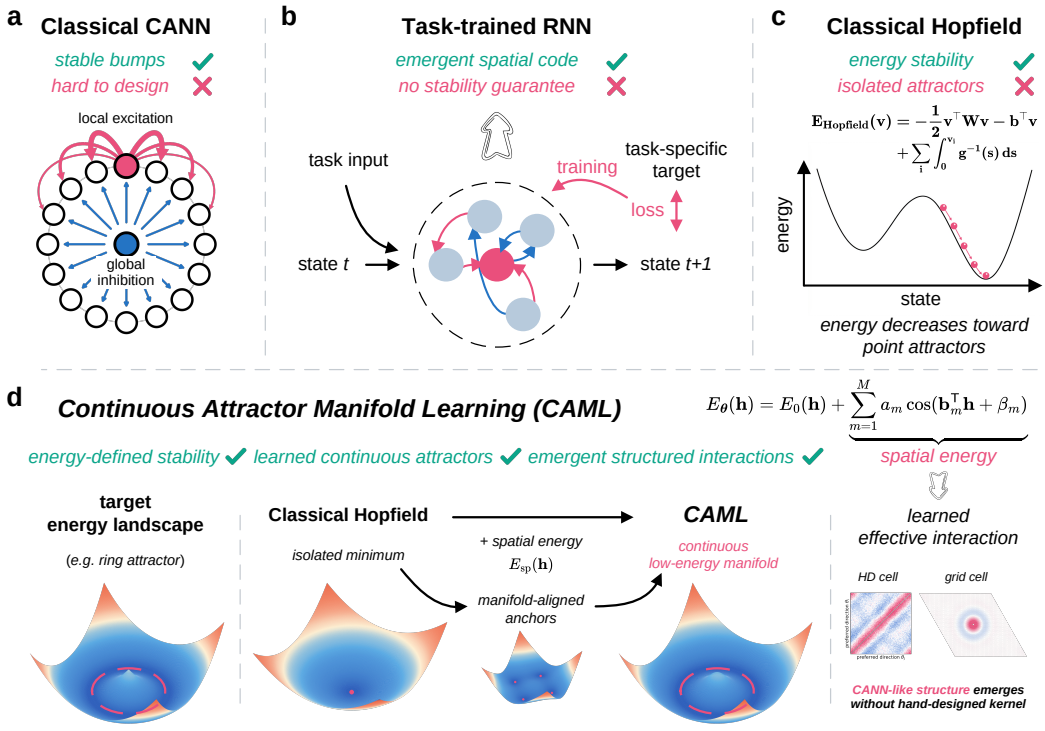


Figure 1: **Overview of Continuous Attractor Manifold Learning (CAML). a**, Classical CANNs stabilize continuous bump states through structured excitation and inhibition, but their recurrent kernels must be hand-designed. **b**, Task-trained RNNs can develop spatial codes from behavioral objectives, yet their learned dynamics lack an explicit stability guarantee. **c**, Classical Hopfield networks provide Lyapunov energy descent, but their attractors are isolated minima rather than continuous spatial manifolds. **d**, CAML learns dynamics that stabilize a continuous spatial manifold by shaping a population energy landscape. The illustrated construction combines a learnable spatial energy with an optional baseline energy. Its gradient supports ring and torus restoration and induces structured effective interactions without a hand-designed recurrent kernel.

Trainable recurrent models provide a complementary route by optimizing spatial dynamics directly from navigation or path-integration objectives. RNNs trained in this way can develop grid-like representations, while neural ODEs offer a flexible framework for learning continuous-time latent dynamics [15, 16, 17, 18]. Yet their learned dynamics are typically understood after training through emergent population geometry, fixed or slow points, and approximate attractor structure [19, 15, 18]. These models demonstrate that spatial codes can arise from optimization, but leave open how to learn restoration toward a target manifold and controlled motion along it within a common theoretical framework.

Hopfield networks ground recurrent stability in an explicit energy function whose autonomous dynamics descend toward attractors [20, 21]. Dense and modern Hopfield networks extend this principle to high-capacity associative memory and continuous-state retrieval, but they are primarily formulated for memory access rather than continuous-time dynamics that stabilize and control neural manifolds [22, 23, 24]. Dynamic Manifold Hopfield Networks move closer to this direction by preserving continuous Hopfield dynamics while learning cue-dependent energy landscapes that reorganize attractor geometry within a single dynamical system [25]. However, these extensions focus on associative recall or cue-dependent reconfiguration of attractor basins, rather than learning continuous spatial attractors that restore off-manifold perturbations and support input-driven motion.

We propose Continuous Attractor Manifold Learning (CAML) to learn the dynamics that make a spatial population manifold an attractor. CAML separates two complementary roles: restoring perturbed activity toward the manifold and moving activity along it in response to self-motion. We

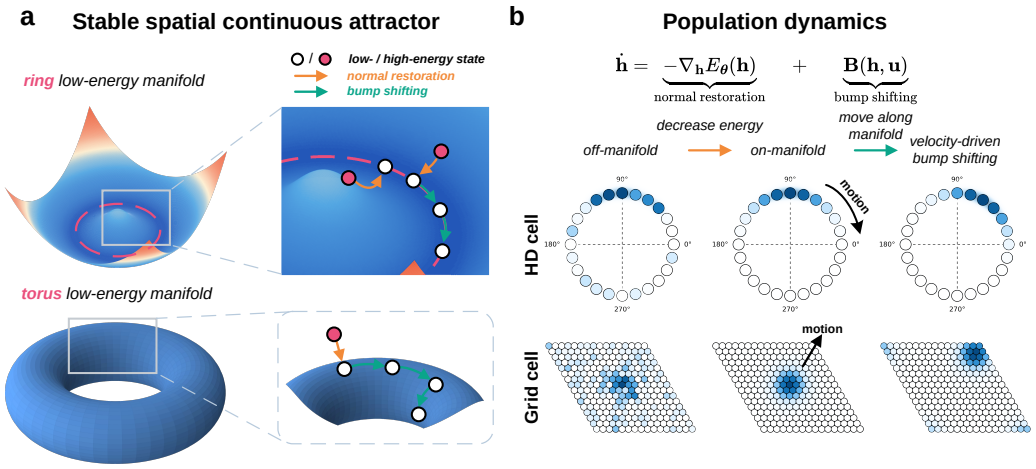


Figure 2: **Energy-based population dynamics of spatial continuous attractors in CAML. a**, CAML learns ring and torus low-energy manifolds in population space for head-direction and grid-cell representations. The energy gradient restores perturbed states toward the manifold, while motion inputs move states along it. **b**, The dynamics decompose as $\dot{\mathbf{h}} = -\nabla_{\mathbf{h}} E_{\boldsymbol{\theta}}(\mathbf{h}) + \mathbf{B}(\mathbf{h}, \mathbf{u})$. The first term provides energy-decreasing normal restoration, and the second provides input-driven bump shifting. This yields stable bump recovery and motion-driven activity translation in both populations.

learn restoration by shaping a scalar energy whose low-energy set represents the target manifold, and learn bump shifting from population changes or trajectories. The population encoding is supplied, so the task is to learn its attractor dynamics rather than discover a representation from raw observations. For head-direction and grid-cell systems, periodic geometry motivates Fourier frequency scales [5, 6, 26, 27]. We embed these scales in full population-space directions, allowing the energy gradient to act on population-level perturbations while velocity-driven bump shifting moves activity along the manifold [11, 8, 10].

The theoretical basis of CAML separates energy descent from the ability to represent a target restoring field. We show that the Fourier construction is an instance of a general Hopfield-compatible energy extension. In this extension, any differentiable feature map with symmetric feature interactions defines an energy-derived restoring force that can be inserted into continuous Hopfield dynamics while preserving Lyapunov descent under standard symmetry and monotonicity conditions [21, 28]. Thus, the feature family determines the expressivity of the learned landscape without altering the energy-descent guarantee. When sufficiently expressive, such landscapes can approximate target attractor energies and their restoring fields [29, 30]. Population energy learning thus provides a route from a target spatial manifold to restoring dynamics with an explicit descent property.

We evaluate CAML on head-direction rings, grid-cell tori, and a bounded, non-periodic place manifold, using both fixed and learned bump shifting. In head-direction and grid-cell systems, the learned ring and torus landscapes recover perturbed states while angular or linear velocity inputs drive population activity along the attractor, matching the restoration and bump shifting requirements of continuous attractor models [7, 12, 8, 11, 10]. We further show that the learned energy induces local effective interactions with ring-like and toroidal structure organized by preferred direction or grid phase, consistent with classical biological attractor models [7, 12, 13, 8, 14, 5, 6]. Together, these results connect learned manifold restoration and bump shifting to interpretable continuous-attractor dynamics for spatial cognition.

Code is available at `https://github.com/birlabs/CAML`.

Our contributions are summarized as follows.

1) **Continuous attractor manifold learning.** We formulate CAML through energy-based restoration and learnable velocity-driven bump shifting.

2) **Theoretical foundation.** We establish autonomous energy descent and conditional approximation of target energies and restoring fields in a general Hopfield-compatible construction.

3) **Spatial dynamics and emergent interactions.** We instantiate CAML with Fourier energies for HD/Grid and combined quadratic and Fourier energy for the main Place model, demonstrate recovery and tracking on head-direction, grid, and bounded place manifolds, and show that structured effective interactions emerge from the learned restoring dynamics.

2 Continuous Attractor Manifold Learning

CAML learns two components of spatial attractor dynamics: restoration toward a population manifold and bump shifting along it. A learned scalar energy defines the restoring field, while a velocity-dependent field controls motion. This decomposition connects energy-based stability to the restoration and motion required by spatial CANNs [21, 7, 8, 10].

2.1 From Point Attractors to Spatial Manifolds

Classical Hopfield dynamics represents memories as isolated attractors of an energy landscape [20, 21]. For a population state $\mathbf{h} \in \mathbb{R}^N$, this view can be written as an energy whose minima form a finite set,

$$\mathcal{A}_{\text{point}} = \left\{ \mathbf{h}_k \in \mathbb{R}^N \;\middle|\; \mathbf{h}_k \text{ is a stable local minimum of } E_{\boldsymbol{\theta}} \right\}_{k=1}^{K}. \tag{1}$$

Such isolated attractors are appropriate for associative recall, but spatial variables such as orientation and location vary continuously.

CAML instead targets a continuous attractor manifold in population space. Let $\mathbf{x} \in \mathcal{S}$ denote the represented coordinate, and let the encoder $\text{enc} : \mathcal{S} \to \mathbb{R}^N$ map it to population activity through the prescribed tuning curves. The target attractor is

$$\mathcal{M} = \left\{ \text{enc}(\mathbf{x}) \in \mathbb{R}^N \;\middle|\; \mathbf{x} \in \mathcal{S} \right\}, \tag{2}$$

where $\mathcal{S} = \mathbb{S}^1$ for head direction, $\mathcal{S} = \mathbb{T}^2$ for grid-cell phase, and a bounded subset of $\mathbb{R}^2$ for the place manifold. The goal is to shape an energy landscape whose low-energy set is $\mathcal{M}$, rather than a finite collection of isolated states.

We model restoration toward the population manifold and bump shifting along it with the dynamics

$$\dot{\mathbf{h}} = \underbrace{-\nabla_{\mathbf{h}} E_{\boldsymbol{\theta}}(\mathbf{h})}_{\text{normal restoration}} + \underbrace{\mathbf{B}(\mathbf{h}, \mathbf{u})}_{\text{bump shifting}}. \tag{3}$$

Here $\mathbf{u}$ collects velocity inputs and $\mathbf{B}$ is the bump-shifting field. The first term restores off-manifold perturbations, while the second moves activity along the manifold to track self-motion [7, 8, 11, 2].

2.2 Parameterizing Manifold Restoration with a Population Energy

We parameterize the restoring field as the negative gradient of a learned population energy. In the general Hopfield-compatible form, the total energy can be written as

$$E_{\boldsymbol{\theta}}(\mathbf{h}) = E_0(\mathbf{h}) + E_{\text{sp}}(\mathbf{h}), \tag{4}$$

Here E_0 denotes an optional quadratic Hopfield energy and E_{sp} is the learned spatial Fourier energy. HD and Grid use $E_0 = 0$, whereas the main Place model combines $E_0(\mathbf{h}) = -\frac{1}{2}\mathbf{h}^\top \mathbf{W}\mathbf{h}$ with the Fourier term, up to additive calibration constants. Fourier-only and quadratic-only Place models are ablations. The more general Continuous Hopfield insertion is derived in Appendix A.

The learned population energy is represented by a finite Fourier expansion,

$$E_{\text{sp}}(\mathbf{h}) = e_0 + \sum_{m=1}^{M} a_m \cos\left(\mathbf{b}_m^\top \mathbf{h} + \beta_m\right), \tag{5}$$

where $\mathbf{b}_m \in \mathbb{R}^N$ is a population-space direction. Taking its negative gradient gives the autonomous restoring dynamics directly:

$$\dot{\mathbf{h}} = \mathbf{F}_{\boldsymbol{\theta}}(\mathbf{h}) = -\nabla_h E_{\boldsymbol{\theta}}(\mathbf{h}) = -\nabla_h E_0(\mathbf{h}) + \sum_{m=1}^{M} a_m \sin(\mathbf{b}_m^\top \mathbf{h} + \beta_m)\mathbf{b}_m. \tag{6}$$

We learn the coefficients a_m and population directions $\mathbf{b}_m$ by gradient optimization, then rescale the force magnitude and center the energy. Appendices B and C give the initialization, optimization, and calibration details.

2.3 Energy Descent and Learning Objective

We train $E_{\boldsymbol{\theta}}$ using clean population states $\mathbf{h}^\star = \mathrm{enc}(\mathbf{x})$ and their perturbed versions $\tilde{\mathbf{h}}$. The decoder $\mathrm{dec} : \mathbb{R}^N \to \mathcal{S}$ estimates the represented coordinate. Composing it with the encoder maps an off-manifold state back to a valid population code:

$$\mathrm{map}_{\mathcal{M}}(\tilde{\mathbf{h}}) = \mathrm{enc}\Big(\mathrm{dec}(\tilde{\mathbf{h}})\Big) \in \mathcal{M}. \tag{7}$$

For HD, dec uses the activity-weighted circular mean, and Grid applies this readout to each phase. The map returns a state on $\mathcal{M}$, not necessarily its Euclidean nearest point. Encoding, decoding, and sampling details are given in Appendix C. The local target energy used for learning is

$$E^\star(\tilde{\mathbf{h}}) = \frac{1}{\alpha_N}\left\|\tilde{\mathbf{h}} - \mathrm{sg}\Big[\mathrm{map}_{\mathcal{M}}(\tilde{\mathbf{h}})\Big]\right\|_2^2, \tag{8}$$

where $\mathrm{sg}[\cdot]$ denotes stop-gradient, treating the mapped state as a fixed target for the current sample, and $\alpha_N = \sqrt{N}$ normalizes the energy scale across population sizes. With the mapped state held fixed by stop-gradient, the negative gradient of this target energy gives the target restoring field:

$$\mathbf{F}^\star(\tilde{\mathbf{h}}) = -\nabla_{\tilde{\mathbf{h}}} E^\star(\tilde{\mathbf{h}}) = \frac{2}{\alpha_N}\left(\mathrm{sg}\Big[\mathrm{map}_{\mathcal{M}}(\tilde{\mathbf{h}})\Big] - \tilde{\mathbf{h}}\right). \tag{9}$$

This target points from the perturbed state toward the decoded population code.

The fitting objective has two roles. First, the learned force should align with the manifold-mapped restoring field:

$$\mathcal{L}_{\mathrm{dir}} = \mathbb{E}_{\tilde{\mathbf{h}}}\left[1 - \cos\angle\Big(\mathbf{F}_{\boldsymbol{\theta}}(\tilde{\mathbf{h}}), \mathbf{F}^\star(\tilde{\mathbf{h}})\Big)\right]. \tag{10}$$

Here $\cos\angle(\mathbf{a}, \mathbf{b})$ denotes the cosine of the angle between two nonzero vectors. Second, we anchor the energy of clean manifold states near zero:

$$\mathcal{L}_{\mathrm{anch}} = \mathbb{E}_{\mathbf{h}^\star \in \mathcal{M}}\left[E_{\boldsymbol{\theta}}(\mathbf{h}^\star)^2\right]. \tag{11}$$

We minimize $\mathcal{L} = \lambda_{\mathrm{dir}}\mathcal{L}_{\mathrm{dir}} + \lambda_{\mathrm{anch}}\mathcal{L}_{\mathrm{anch}}$ by gradient training. Training states combine fresh perturbations with detached rollouts of the current restoring dynamics. Appendix C gives the sampling and optimization details.

Energy fitting

1. Randomly initialize the Fourier basis and any quadratic weights. Set $a_m = e_0 = 0$.
2. Sample clean states, fresh perturbations, and detached restoring trajectories.
3. Optimize a_m, $\mathbf{b}_m$, and $\mathbf{W}$ when present using $\mathcal{L}$, periodically refreshing trajectory samples.
4. Jointly rescale the force components and center their clean-state energies with additive offsets. These offsets do not affect dynamics.

The advantage of learning a scalar energy is that autonomous restoration has an explicit Lyapunov structure. When the bump-shifting term in Eq. (3) is absent, the dynamics reduce to energy descent,

$$\dot{\mathbf{h}} = -\nabla_{\mathbf{h}} E_{\boldsymbol{\theta}}(\mathbf{h}), \qquad \frac{dE_{\boldsymbol{\theta}}}{dt} = -\left\|\nabla_{\mathbf{h}} E_{\boldsymbol{\theta}}(\mathbf{h})\right\|_2^2 \le 0. \tag{12}$$

Thus, autonomous restoration follows an energy-decreasing flow. With bump shifting, the energy rate includes the additional term $\nabla_{\mathbf{h}} E_{\boldsymbol{\theta}}(\mathbf{h})^\top \mathbf{B}(\mathbf{h}, \mathbf{u})$. More generally, the spatial energy in Eq. (4) can be inserted into continuous Hopfield dynamics while preserving energy descent under the standard symmetry and monotonicity conditions [21, 28]. The full derivation is provided in Appendix A.

2.4 Fixed and Learned Bump Shifting Along the Manifold

Autonomous restoration does not provide controlled motion along the represented variable. Spatial tracking therefore requires a velocity-driven bump-shifting term in Eq. (3). We first describe fixed operators on periodic manifolds, then their learned counterparts. For head direction, self-motion provides an angular velocity input ω, and the population bump is shifted along the ring by

$$\mathbf{B}_{\mathrm{HD}}(\mathbf{h}, \omega) = -\omega \mathbf{D}_\theta \mathbf{h}, \tag{13}$$

where $\mathbf{D}_\theta$ is a periodic finite-difference operator over preferred directions. The full HD dynamics are

$$\dot{\mathbf{h}} = -\nabla_{\mathbf{h}} E_{\boldsymbol{\theta}}(\mathbf{h}) - \omega \mathbf{D}_\theta \mathbf{h}. \tag{14}$$

The first term restores the bump to the learned ring manifold, while the second term shifts it according to angular velocity [7, 12, 11].

For grid cells, the represented state has two periodic phases. Let ω_x and ω_y denote the velocities along its two phase directions, induced by planar motion. Using periodic finite-difference operators $\mathbf{D}_x$ and $\mathbf{D}_y$, the bump-shifting term is

$$\mathbf{B}_{\text{grid}}(\mathbf{h}, \omega_x, \omega_y) = -\omega_x \mathbf{D}_x \mathbf{h} - \omega_y \mathbf{D}_y \mathbf{h}, \tag{15}$$

and the driven dynamics become

$$\dot{\mathbf{h}} = -\nabla_{\mathbf{h}} E_{\boldsymbol{\theta}}(\mathbf{h}) - \omega_x \mathbf{D}_x \mathbf{h} - \omega_y \mathbf{D}_y \mathbf{h}. \tag{16}$$

This separates stable recovery from path integration: the learned energy maintains the toroidal population manifold, while velocity inputs translate activity along its two phase directions [8, 13, 14].

To learn bump shifting, we use $\mathbf{B}_{\boldsymbol{\phi}}(\mathbf{h}, \mathbf{u}) = \sum_j u_j \mathbf{G}_{\boldsymbol{\phi},j}(\mathbf{h})$, keeping the fitted energy fixed. For HD/Grid, linear operators are fitted to local changes of population codes. For the place manifold, a population-state MLP is trained by matching multi-step trajectories. CAML thus learns restoration and bump shifting in separate stages. Appendix F gives the training details.

3 Learned Continuous Attractor Manifolds for Spatial Cognition

3.1 Restoration to Ring and Torus Manifolds

A spatial continuous attractor must recover off-manifold population states while preserving the represented variable. We test this property without bump shifting by perturbing clean HD and grid-cell population states and evolving them under the autonomous restoring dynamics in Eq. (6). For the visualization in Fig. 3, recovery is measured by the manifold-mapped population distance

$$\bar{d} = \frac{1}{Q} \sum_{q=1}^{Q} \left\| \mathbf{h}_q - \text{map}_{\mathcal{M}}(\mathbf{h}_q) \right\|_2, \tag{17}$$

where $\text{map}_{\mathcal{M}}$ is the map obtained by decoding and re-encoding used in Eq. (7). We further quantify whether the learned energy matches the target restoration field by

$$\text{FA} = \frac{1}{|\mathcal{A}|} \sum_{q \in \mathcal{A}} \cos\angle\Big(\mathbf{F}_{\boldsymbol{\theta}}(\tilde{\mathbf{h}}_q), \mathbf{F}^\star(\tilde{\mathbf{h}}_q) \Big), \qquad \mathcal{A} = \{q : \|\mathbf{F}^\star(\tilde{\mathbf{h}}_q)\|_2 > 10^{-4}\}, \tag{18}$$

with $\mathbf{F}_{\boldsymbol{\theta}}(\mathbf{h}) = -\nabla_{\mathbf{h}} E_{\boldsymbol{\theta}}(\mathbf{h})$ and $\mathbf{F}^\star$ defined in Eq. (9). Energy ordering is measured by Spearman's rank correlation [31]

$$\rho_S = \text{corr}_{\text{P}} \left(\text{rank}\{E_{\boldsymbol{\theta}}(\tilde{\mathbf{h}}_q)\}_{q=1}^{Q}, \text{rank}\{E^\star(\tilde{\mathbf{h}}_q)\}_{q=1}^{Q} \right), \tag{19}$$

where $E^\star$ is the manifold-mapped target energy in Eq. (8). FA and energy correlation use 20,000 independently perturbed states with absolute Gaussian SD sampled uniformly from $[0.05, 0.5]$.

For HD cells, noisy population states form a dispersed annulus in a fixed linear circular embedding of raw activity. Energy-based restoration concentrates these states near the ring and reduces the mean mapping distance from 2.426 to 0.014 after 25 s (Fig. 3a,b). The learned force aligns with the manifold-mapped target with $\text{FA} = 0.969$, and the learned energy preserves the target distance ordering with $\rho_S = 0.998$. The activity snapshots show recovery of a localized bump, while Fig. 3c separately measures error to the original phase.

Grid-cell populations show the same restoration principle on a toroidal manifold. Perturbed states concentrate near the toroidal code manifold, reducing the mean mapping distance from 32.071 to 0.350 after 200 s (Fig. 3d,e). The learned grid energy reaches $\text{FA} = 0.988$ and $\rho_S > 0.999$. Population snapshots show recovery of a localized bump on the rhombic lattice. Residual phase

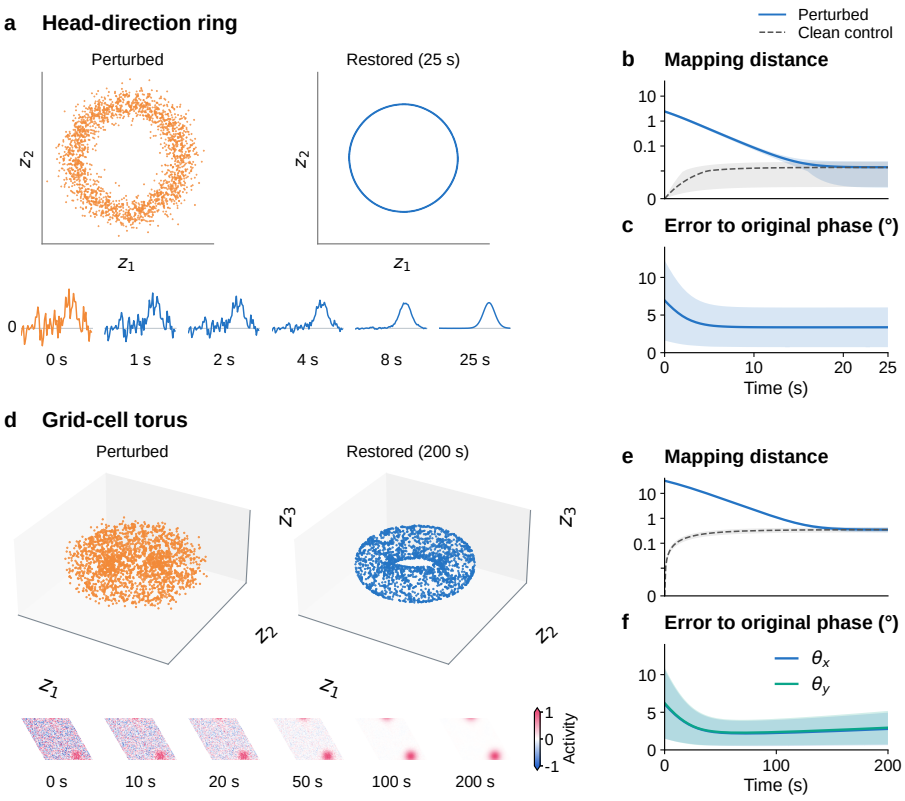


Figure 3: **Learned population energies organize neural population activity into ring and torus manifolds. a,d**, HD and Grid population embeddings and activity snapshots before and after restoration. **b,e**, Distance to the mapped population code. Dashed curves show clean controls. **c,f**, Absolute wrapped error to the original phase. Curves show mean $\pm$ SD.

error remains (Fig. 3f), so recovery toward the manifold does not imply exact recovery of the original coordinate.

These results support the restoration component of CAML. The learned scalar energy brings perturbed states close to the supplied manifolds before any bump shifting is applied. Clean controls reach similarly small mapping distances (HD: 0.014, Grid: 0.350), indicating a finite-model residual rather than an exact fixed manifold.

For recovery comparisons, we compute population RMSE for each test state relative to its original clean activity, normalize by the reference RMS, and report the mean and SD across states. We evaluate CAML recovery after 25 s (HD) or 200 s (Grid). Table 1 uses identical prescribed codes and Gaussian perturbations for CAML, the DMHN-derived baseline [25], and iterative attention-based Modern Hopfield retrieval (MHN) [23]. CAML has lower error than these two shared-code baselines in all four conditions shown. For HD/Grid CANNs [7, 8], clean reference states are generated by their preparatory dynamics rather than by the prescribed encoding functions. Their recovery errors therefore also include drift of these reference states. Appendix E.1 gives the metric definition and baseline settings.

Across the higher HD noise levels $\sigma = 0.5, 1, 2$, CAML reduces mean recovery error by 41–50% relative to direct decoding. The comparison also depends on noise structure. Under Grid quadrant bias of strength 0.5, CAML gives error 0.086, versus 0.357 for decoding and re-encoding. Appendix E.2 reports the full noise sweeps.

Table 1: **Noise recovery.** Normalized population RMSE ↓, mean ± SD across test states. Gaussian noise SD is σ times the reference RMS. Bold marks the lowest mean in each row.

Population	σ	CAML	DMHN-derived	MHN	CANN†
HD	0.1	$\mathbf{0.014 \pm 0.009}$	0.474 ± 0.046	1.093 ± 0.001	0.120 ± 0.010
HD	0.5	$\mathbf{0.051 \pm 0.038}$	0.482 ± 0.060	1.093 ± 0.001	0.155 ± 0.054
Grid	0.1	$\mathbf{0.078 \pm 0.032}$	1.000 ± 0.000	0.995 ± 0.000	0.153 ± 0.114
Grid	0.5	$\mathbf{0.079 \pm 0.032}$	1.000 ± 0.000	0.995 ± 0.000	0.156 ± 0.116

† CANN uses its own reference states. Other methods use prescribed codes.

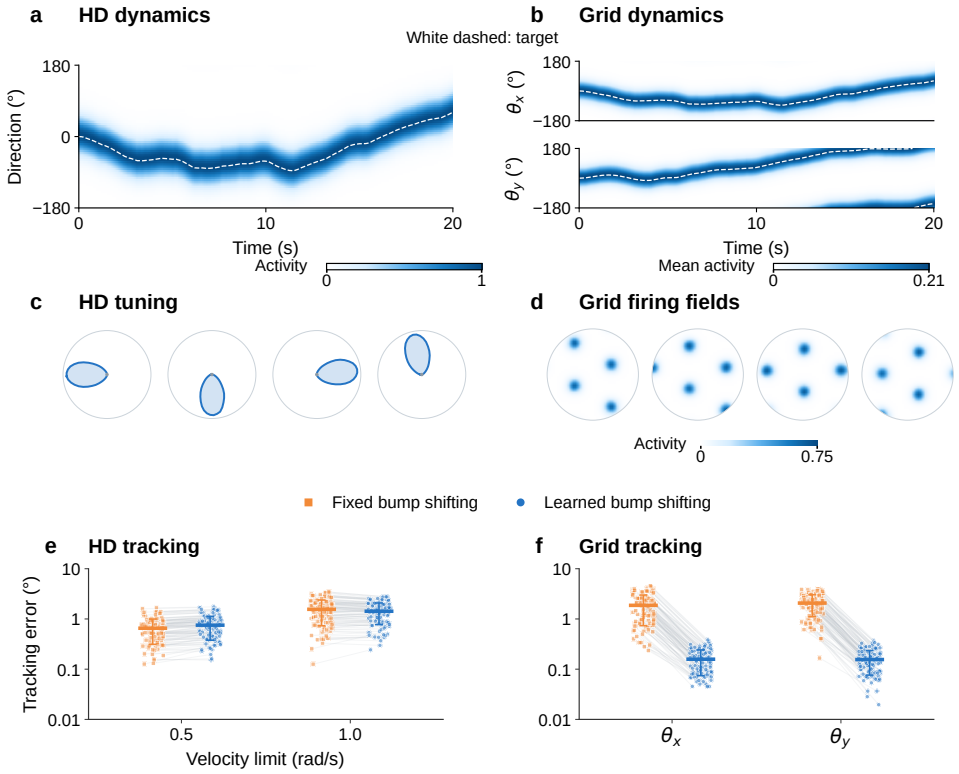


Figure 4: **Velocity-driven dynamics and tracking. a,b**, HD and Grid activity during fixed bump shifting. White dashed lines show target phases. **c,d**, Directional tuning and spatial firing maps for example neurons. **e,f**, Fixed versus learned tracking errors for HD and Grid. Points show 100 paired trajectories with mean ± SD.

3.2 Bump Shifting Supports Tracking on Periodic and Bounded Manifolds

Energy descent restores corrupted states to the attractor, but spatial coding also requires controlled motion along the represented variable. We therefore test the driven dynamics in Eqs. (14) and (16), where the learned energy supplies normal stabilization and fixed finite-difference operators supply bump shifting. For HD cells, the input is angular velocity $\omega(t)$. For grid cells, a planar velocity $\mathbf{v}(t) \in \mathbb{R}^2$ is mapped to torus phase velocities by

$$\begin{bmatrix} \omega_x(t) \\ \omega_y(t) \end{bmatrix} = s \begin{bmatrix} 1 & 0 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \mathbf{v}(t), \tag{20}$$

where s sets the phase scale. Equation (20) maps physical velocities to Grid phase velocities for the spatial firing maps in Fig. 4d, using $s = 2\,\mathrm{rad/m}$. The paired benchmark and Fig. 4a,b use angular or phase inputs from a bounded random-walk velocity process. Random-walk velocities have Gaussian increments with SD $0.3\sqrt{dt}$ and are bounded by the prescribed velocity limit. Grid uses independent increments for its two phase axes, with the same increment distribution and per-axis velocity limit.

Table 2: **Fixed versus learned bump shifting.** Tracking error ↓, mean ± SD over 100 paired trajectories. Grid errors average the two phase axes. Bold marks the lower mean.

Population	Velocity setting	CAML (learned)	CAML (fixed)
HD	0.5 rad/s limit	$0.752 \pm 0.375^\circ$	$\mathbf{0.654 \pm 0.342}^\circ$
HD	1.0 rad/s limit	$\mathbf{1.425 \pm 0.653}^\circ$	$1.557 \pm 0.825^\circ$
Grid	0.5 rad/s limit	$\mathbf{0.093 \pm 0.037}^\circ$	$0.715 \pm 0.287^\circ$
Grid	1.0 rad/s limit	$\mathbf{0.157 \pm 0.050}^\circ$	$1.968 \pm 0.731^\circ$
Place	Random walk	$\mathbf{0.0039 \pm 0.0023}\,\mathrm{m}$	$0.0671 \pm 0.0700\,\mathrm{m}$

For HD cells, random-walk angular velocity moves a localized bump around the ring (Fig. 4a). The accumulated tuning curves remain direction selective (Fig. 4c). The restoring field stabilizes the moving bump as the bump-shifting term controls its angular displacement.

Grid-cell populations show the analogous behavior on a torus. Under random-walk inputs, the bump follows both torus phases (Fig. 4b). The same dynamics produce spatially periodic firing maps in the arena (Fig. 4d). These maps arise from velocity-driven bump shifting on the learned toroidal manifold.

We next replace fixed bump shifting with learned bump shifting while keeping the energy unchanged (Table 2, Fig. 4e,f). HD uses $N = 64, M = 4096$ and 20 s trajectories. Grid uses $N = 64^2, M = 16384$ and 40 s trajectories. Each velocity condition uses 100 paired random walks, excluding the first 10% of each trajectory. Fixed HD shifting has lower mean error at the 0.5 rad/s limit, while learned shifting has lower mean error at 1.0 rad/s. Learned Grid shifting reduces the mean error averaged over phase axes by 87% at the 0.5 rad/s per-axis limit and 92% at 1.0 rad/s.

For a non-periodic example, $N = 256$ localized Gaussian tuning curves encode position in a bounded $1\,\mathrm{m} \times 1\,\mathrm{m}$ arena, with $M = 8192$ Fourier features. The main Place model uses the sum of a quadratic Hopfield energy and the learned Fourier energy. Clean training and recovery positions satisfy $x, y \in [0.1, 0.9]\,\mathrm{m}$, leaving a $0.1\,\mathrm{m}$ margin along the arena boundary. With absolute activity-noise SD 0.3, recovery MSE is $(1.60 \pm 1.62) \times 10^{-3}$ across 20,000 test states. Over 100 paired $20\,\mathrm{s}$ random walks, learned bump shifting gives mean position error $0.0039 \pm 0.0023\,\mathrm{m}$, compared with $0.0671 \pm 0.0700\,\mathrm{m}$ for fixed shifting. Both receive the same velocity inputs and use the same bounded-map decoder. Appendix F.2 gives the Place training and evaluation settings and energy ablation.

3.3 Continuous-Attractor Interaction Structure Emerges from Learning

Classical continuous attractor models impose structured recurrent weights to stabilize activity bumps on rings or tori [32, 7, 8]. CAML does not prescribe such weights. To test whether an analogous structure emerges from energy learning, we linearize the learned restoring field around clean population states,

$$\mathbf{F}_{\boldsymbol{\theta}}(\mathbf{h}_0 + \delta\mathbf{h}) \approx \mathbf{F}_{\boldsymbol{\theta}}(\mathbf{h}_0) + \mathbf{J}_{\boldsymbol{\theta}}(\mathbf{h}_0)\delta\mathbf{h}, \qquad \mathbf{J}_{\boldsymbol{\theta}}(\mathbf{h}_0) = \left.\frac{\partial \mathbf{F}_{\boldsymbol{\theta}}(\mathbf{h})}{\partial \mathbf{h}}\right|_{\mathbf{h}=\mathbf{h}_0}, \tag{21}$$

where $\mathbf{F}_{\boldsymbol{\theta}}(\mathbf{h}) = -\nabla_{\mathbf{h}} E_{\boldsymbol{\theta}}(\mathbf{h})$. For the Fourier energy in Eq. (5), this gives

$$\mathbf{J}_{\boldsymbol{\theta}}(\mathbf{h}_0) = \sum_{m=1}^{M} a_m \cos(\mathbf{b}_m^\top \mathbf{h}_0 + \beta_m)\mathbf{b}_m \mathbf{b}_m^\top. \tag{22}$$

We interpret $\mathbf{J}_{\boldsymbol{\theta}}(\mathbf{h}_0)$ as a local effective interaction induced by the learned scalar energy. It is not an explicitly parameterized synaptic matrix. Appendix D describes how the mean and state-dependent interaction profiles are computed.

For HD cells, averaging $\mathbf{J}_{\boldsymbol{\theta}}(\mathbf{h}_0)$ over sampled bump positions yields a nearly circulant interaction matrix organized by preferred direction (Fig. 5a). The corresponding static profile $W(\Delta\theta)$ has a local excitatory peak near $\Delta\theta = 0$ and inhibitory side lobes, matching the qualitative form of hand-designed ring-attractor kernels. The instantaneous interaction is not fixed, however. Its dependence on bump position is summarized by aligning the interactions to the active bump centre (Fig. 5b).

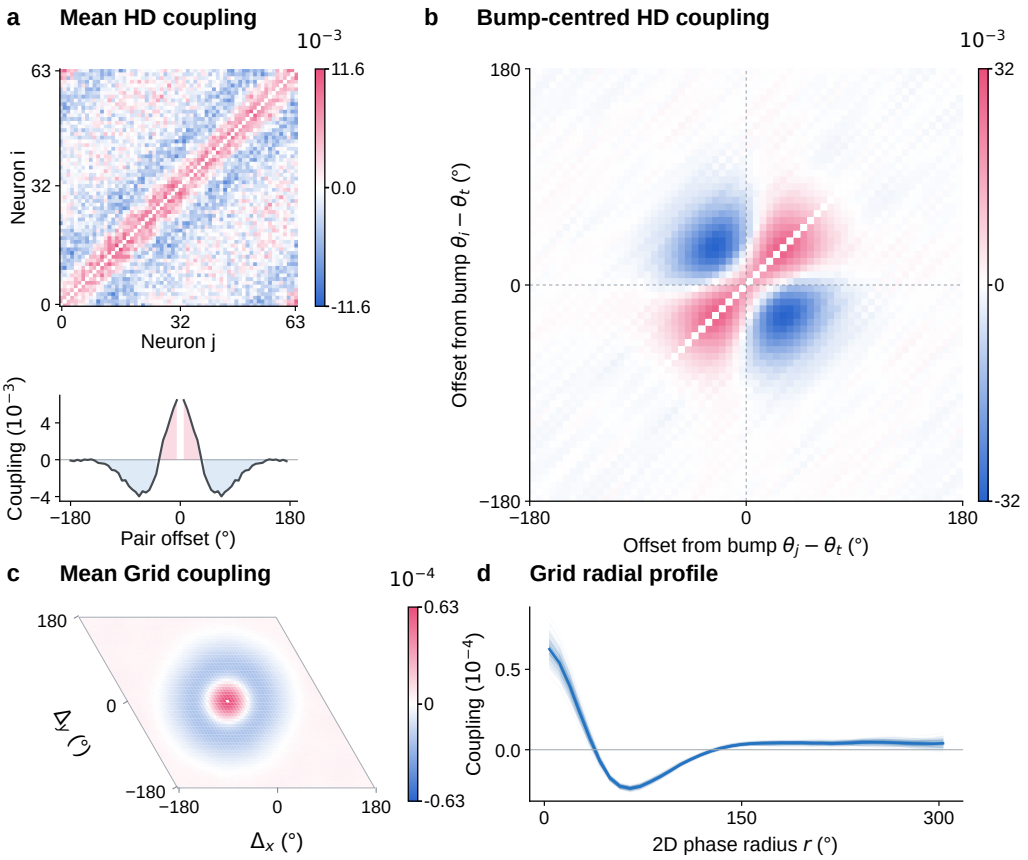


Figure 5: **Continuous-attractor interaction structure emerges from the learned energy. a**, Mean HD Jacobian and its profile over neuron-pair offsets. **b**, HD interactions aligned to the active bump centre. **c**, Mean Grid coupling kernel in hexagonal offset coordinates. **d**, Radial Grid coupling profiles: mean (thick curve) and individual bump phases (thin curves). Self-interactions are excluded. Analysis details are given in Appendix D.

Thus the learned HD energy induces a state-dependent interaction rather than a single hand-designed recurrent kernel.

Grid cells show an analogous center-surround structure. After averaging over sampled on-manifold grid bumps, the learned interaction profile forms a localized center-surround pattern in hexagonal coordinates (Fig. 5c). The radial profile has a local excitatory peak and broader inhibitory surround (Fig. 5d), qualitatively resembling the Mexican-hat kernels used in continuous attractor models.

CANN-like local interactions therefore emerge in the Jacobian of the learned restoring field.

4 Conclusion

CAML provides a framework for learning the restoration and bump-shifting dynamics of spatial continuous attractor manifolds. Population energy learning supplies autonomous restoration, and learned bump shifting extends these dynamics to velocity-driven tracking on HD, Grid, and bounded place manifolds. The emergent local interaction structure connects this learned dynamical description to classical CANN mechanisms. The approach assumes supplied population encodings and restoration targets. Its recovery advantage depends on noise structure and amplitude, as direct decoding is more accurate under weak Grid Gaussian noise. The energy-descent guarantee applies to autonomous restoration and does not cover arbitrary velocity-driven dynamics. Future work should extend CAML to jointly learn representations and sensory anchoring, scale to heterogeneous multi-module codes, and test spatial remapping and episodic memory dynamics in more complex environments.

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A Hopfield Energy Descent and Expressivity

A.1 Energy-Extended Hopfield Dynamics

Let $\mathbf{h} \in \mathbb{R}^N$ denote neural activity. In the general continuous Hopfield model, $\mathbf{h} = \mathbf{g}(\boldsymbol{\xi})$, where $\boldsymbol{\xi}$ is the internal potential. Let $\mathbf{T} = \mathrm{diag}(\tau_i)$ with $\tau_i > 0$, and let $\mathbf{b}_\mathrm{H}$ be a constant bias. For symmetric $\mathbf{W}_\mathrm{H}$ and differentiable, strictly increasing g_i, the continuous Hopfield energy is

$$E_\mathrm{H}(\mathbf{h}) = -\tfrac{1}{2}\mathbf{h}^\top \mathbf{W}_\mathrm{H}\mathbf{h} - \mathbf{b}_\mathrm{H}^\top \mathbf{h} + \sum_i \int_0^{h_i} g_i^{-1}(s)\, ds. \tag{23}$$

Adding a continuously differentiable term E_sp gives $E = E_\mathrm{H} + E_\mathrm{sp}$ and the autonomous dynamics

$$\mathbf{T}\dot{\boldsymbol{\xi}} = -\boldsymbol{\xi} + \mathbf{W}_\mathrm{H}\mathbf{h} + \mathbf{b}_\mathrm{H} - \nabla_\mathbf{h} E_\mathrm{sp}(\mathbf{h}). \tag{24}$$

A.2 Energy Descent

Theorem 1 (Energy descent)**.** *If $\mathbf{W}_\mathrm{H} = \mathbf{W}_\mathrm{H}^\top$, $\tau_i > 0$, and $g_i'(\xi_i) > 0$, the dynamics in Eq. (24) satisfy*

$$\frac{dE}{dt} = -\sum_i \tau_i g_i'(\xi_i)(\dot{\xi}_i)^2 \le 0. \tag{25}$$

Proof. Since $\mathbf{g}^{-1}(\mathbf{h}) = \boldsymbol{\xi}$, differentiation gives $\nabla_\mathbf{h} E = \boldsymbol{\xi} - \mathbf{W}_\mathrm{H}\mathbf{h} - \mathbf{b}_\mathrm{H} + \nabla_\mathbf{h} E_\mathrm{sp} = -\mathbf{T}\dot{\boldsymbol{\xi}}$. Using $\dot{h}_i = g_i'(\xi_i)\dot{\xi}_i$ in $dE/dt = (\nabla_\mathbf{h} E)^\top \dot{\mathbf{h}}$ yields Eq. (25). □

In CAML, neural activity $\mathbf{h}$ is the sole dynamical state, corresponding to the identity activation $\mathbf{g}(\boldsymbol{\xi}) = \boldsymbol{\xi}$ and hence $\mathbf{h} = \boldsymbol{\xi}$. The autonomous dynamics directly follow $\dot{\mathbf{h}} = -\nabla_\mathbf{h} E_{\boldsymbol{\theta}}(\mathbf{h})$ for the energy in Eq. (4), with $dE_{\boldsymbol{\theta}}/dt = -\|\nabla_\mathbf{h} E_{\boldsymbol{\theta}}\|_2^2$.

A.3 Feature Interactions

For a differentiable feature map $\boldsymbol{\phi} : \mathbb{R}^N \to \mathbb{R}^K$ and symmetric $\mathbf{P}$, choose

$$E_\mathrm{sp}(\mathbf{h}) = -\tfrac{1}{2}\boldsymbol{\phi}(\mathbf{h})^\top \mathbf{P}\boldsymbol{\phi}(\mathbf{h}), \qquad -\nabla_\mathbf{h} E_\mathrm{sp} = \mathbf{J}_{\boldsymbol{\phi}}(\mathbf{h})^\top \mathbf{P}\boldsymbol{\phi}(\mathbf{h}), \tag{26}$$

where $\mathbf{J}_{\boldsymbol{\phi}} = \partial\boldsymbol{\phi}/\partial\mathbf{h}$. Inserting this force into Eq. (24) recovers feature-interaction Hopfield dynamics and preserves Theorem 1.

A.4 Fourier Energy as a Feature-Interaction Kernel

Set $c_m(\mathbf{h}) = \cos(\mathbf{b}_m^\top \mathbf{h} + \beta_m)$ and $\boldsymbol{\Phi}(\mathbf{h}) = (1, c_1(\mathbf{h}), \ldots, c_M(\mathbf{h}))^\top$. The symmetric matrix with $P_{00} = -2e_0$, $P_{0m} = P_{m0} = -a_m$, and all other entries zero satisfies

$$-\tfrac{1}{2}\boldsymbol{\Phi}(\mathbf{h})^\top \mathbf{P}\boldsymbol{\Phi}(\mathbf{h}) = e_0 + \sum_{m=1}^{M} a_m \cos(\mathbf{b}_m^\top \mathbf{h} + \beta_m). \tag{27}$$

Thus CAML's Fourier energy is a particular symmetric feature-interaction energy.

A.5 Bump Shifting and Expressivity

With $\mathbf{u}$ denoting the external velocity input, the driven dynamics $\dot{\mathbf{h}} = -\nabla_\mathbf{h} E_{\boldsymbol{\theta}}(\mathbf{h}) + \mathbf{B}(\mathbf{h}, \mathbf{u})$ satisfy

$$\frac{dE_{\boldsymbol{\theta}}}{dt} = -\|\nabla_\mathbf{h} E_{\boldsymbol{\theta}}\|_2^2 + (\nabla_\mathbf{h} E_{\boldsymbol{\theta}})^\top \mathbf{B}. \tag{28}$$

The autonomous descent guarantee extends to driven dynamics when the second term vanishes.

Expressivity depends on the chosen energy family. If it is dense in $C^1(\Omega)$ on a compact domain, then any target $E_\mathrm{target} \in C^1(\Omega)$ can be approximated uniformly together with its gradient [29, 30]. This is a conditional approximation statement, rather than a guarantee for a fixed finite number of Fourier modes.

B Population Fourier Energy Bases

The Fourier energy and its restoring field are defined in Eqs. (5) and (6). Each direction is initialized as $\mathbf{b}_m = r_m \mathbf{q}_m$, with $\mathbf{q}_m$ sampled uniformly on $\mathbb{S}^{N-1}$ and r_m selected uniformly from 32 logarithmically spaced frequency shells between 0.5 and 8. Phases $\beta_m = 0$ remain fixed. We initialize $a_m = e_0 = 0$ and learn a_m and $\mathbf{b}_m$.

C Energy and Force Matching

Training minimizes the force-direction and clean-energy objectives in Eqs. (10) and (11).

C.1 Training Samples

Clean states are sampled as $\mathbf{h}^\star = \mathrm{enc}(\mathbf{x}) \in \mathcal{M}$, with HD headings and Grid phases drawn uniformly on their periodic domains. Perturbed states are generated by population-space noise with random amplitude,

$$\tilde{\mathbf{h}} = \mathrm{enc}(\mathbf{x}) + \sigma\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \qquad \sigma \sim \mathcal{U}(\sigma_{\min}, \sigma_{\max}). \tag{29}$$

We use absolute noise amplitudes $\sigma \in [0.05, 0.5]$. Every 1000 optimizer updates, we refresh a replay pool from 128 perturbed initial states, collecting detached states after 10, 20, 50, 100, and 200 s of dynamics with $dt = 0.05$ s. Each training batch contains fresh clean states and their noisy counterparts; replay states replace half of the directional-loss samples. The clean batch supplies the energy anchor, and targets are recomputed at each sampled state.

C.2 Population Encoding and Decoding

The targets in Eqs. (8) and (9) use the following population encoders and decoders.

For HD cells, preferred directions θ_i are evenly spaced on $[-\pi, \pi)$, and the population code is

$$\mathrm{enc}_{\mathrm{HD},i}(\theta) = \exp[\kappa \cos(\theta - \theta_i) - \kappa]\,. \tag{30}$$

Given a population state $\mathbf{h}$, we decode

$$\mathrm{dec}_{\mathrm{HD}}(\mathbf{h}) = \mathrm{atan2}\left(\sum_i h_i \sin\theta_i, \sum_i h_i \cos\theta_i\right). \tag{31}$$

For grid cells, preferred phases (μ_{ix}, μ_{iy}) form a uniform 64×64 grid on $[-\pi, \pi)^2$. The population code is indexed by two phases $(\theta_x, \theta_y) \in \mathbb{T}^2$:

$$\mathrm{enc}_{\mathrm{grid},i}(\theta_x, \theta_y) = \exp[\kappa\,(\cos d_{ix} + \cos d_{iy} + \cos(d_{iy} - d_{ix}) - 3)]\,, \tag{32}$$

where

$$d_{ix} = \mathrm{wrap}(\theta_x - \mu_{ix}), \qquad d_{iy} = \mathrm{wrap}(\theta_y - \mu_{iy}). \tag{33}$$

Here wrap maps angular differences to $[-\pi, \pi)$. Given a population state $\mathbf{h}$, we decode each phase by population-vector readout,

$$\mathrm{dec}_{\mathrm{grid},k}(\mathbf{h}) = \mathrm{atan2}\left(\sum_i h_i \sin\mu_{ik}, \sum_i h_i \cos\mu_{ik}\right), \qquad k \in \{x, y\}. \tag{34}$$

C.3 Initialization and Gradient Training

We minimize the force-direction and clean-energy losses in Eqs. (10) and (11), using the initialization in Appendix B. We optimize a_m and $\mathbf{b}_m$, keeping the phases and $e_0 = 0$ fixed, with $\lambda_{\mathrm{dir}} = 1$ and $\lambda_{\mathrm{anch}} = 0.5$. Norms in the cosine loss are floored at 10^{-8}, and samples with target-force norm below 10^{-4} are excluded from direction matching. AdamW uses learning rate 3×10^{-4}, weight decay 10^{-5}, gradient-norm clipping at 5, and cosine decay to 5% of the initial learning rate. Batch sizes are 512 (HD), 256 (Grid), and 8192 (Place). We train for 10,000 updates on HD and Place, and 12,000 on Grid. For models with a quadratic component, $\mathbf{W}$ is optimized jointly with the same force direction loss and energy anchor on clean states. It starts from a symmetrized Gaussian draw scaled by $0.01/\sqrt{N}$. Before each optimizer update, we symmetrize $\mathbf{W}$ and constrain its diagonal entries to be nonpositive.

C.4 Analytic Rescaling

Because the direction loss is scale invariant, we then analytically match the force magnitude:

$$c = \frac{\mathbb{E}_{\tilde{h}} \|\mathbf{F}^\star(\tilde{\mathbf{h}})\|_2}{\mathbb{E}_{\tilde{h}} \|\mathbf{F}_{\boldsymbol{\theta}}(\tilde{\mathbf{h}})\|_2}, \qquad a_m \leftarrow c a_m, \qquad \mathbf{W} \leftarrow c\mathbf{W} \quad \text{when present.} \tag{35}$$

Additive constants then center the Fourier and quadratic component energies on clean samples. These offsets do not affect the dynamics.

D Effective Interaction Analysis

For the Fourier-only HD/Grid models, the local interaction at a clean state $\mathbf{h}_0$ is the Jacobian $\mathbf{J}_{\boldsymbol{\theta}}(\mathbf{h}_0)$ in Eq. (22). We summarize this interaction over bump positions and neuron-pair offsets.

For HD cells, we evaluate $\mathbf{J}_{\boldsymbol{\theta}}(\mathrm{enc}_{\mathrm{HD}}(\theta_t))$ over bump positions θ_t. The static interaction is

$$\bar{\mathbf{J}}_{\mathrm{HD}} = \mathbb{E}_{\theta_t} \left[\mathbf{J}_{\boldsymbol{\theta}}(\mathrm{enc}_{\mathrm{HD}}(\theta_t)) \right]. \tag{36}$$

The one-dimensional profile is obtained by averaging entries with the same circular offset,

$$W_{\mathrm{HD}}(\Delta\theta) = \mathbb{E}_{i,j:\theta_i - \theta_j = \Delta\theta} \left[[\bar{\mathbf{J}}_{\mathrm{HD}}]_{ij} \right]. \tag{37}$$

For Fig. 5b, we align each Jacobian to its bump centre and average over the 64 preferred headings. The displayed axes are the circular offsets $\theta_j - \theta_t$ (horizontal) and $\theta_i - \theta_t$ (vertical).

For grid cells, we evaluate $\mathbf{J}_{\boldsymbol{\theta}}(\mathrm{enc}_{\mathrm{grid}}(\theta_{x,t}, \theta_{y,t}))$ over torus phases. The static interaction is

$$\bar{\mathbf{J}}_{\mathrm{grid}} = \mathbb{E}_{\theta_{x,t}, \theta_{y,t}} \left[\mathbf{J}_{\boldsymbol{\theta}}(\mathrm{enc}_{\mathrm{grid}}(\theta_{x,t}, \theta_{y,t})) \right]. \tag{38}$$

For cells indexed by preferred phases (μ_{ix}, μ_{iy}), we average entries with the same torus offset,

$$W_{\mathrm{grid}}(\Delta_x, \Delta_y) = \mathbb{E}_{i,j:\mu_{ix} - \mu_{jx} = \Delta_x,\, \mu_{iy} - \mu_{jy} = \Delta_y} \left[[\bar{\mathbf{J}}_{\mathrm{grid}}]_{ij} \right]. \tag{39}$$

For visualization, torus offsets are mapped to hexagonal plotting coordinates,

$$x = \Delta_x - \frac{1}{2}\Delta_y, \qquad y = \frac{\sqrt{3}}{2}\Delta_y, \qquad r = \sqrt{x^2 + y^2}. \tag{40}$$

The radial profile $W_{\mathrm{grid}}(r)$ is obtained by binning $W_{\mathrm{grid}}(\Delta_x, \Delta_y)$ by r. Offsets $\Delta_x, \Delta_y \in [-180^\circ, 180^\circ)$ define the displayed phase-offset domain. The kernel in Fig. 5c and the thick radial curve in d average over 256 randomly sampled bump phases. Thin curves show state-specific profiles from a separate Cartesian 16×16 grid of phases. Self-interactions are omitted from the displayed kernels and profiles.

E Recovery Evaluation

Experiments used an NVIDIA RTX A4500 GPU with 20 GB of memory.

E.1 Metric and Baseline Settings

We use PyTorch, NumPy, and SciPy (BSD-3-Clause), and Matplotlib (Matplotlib license). We implemented the baselines from the cited formulations. The FieteLab MATLAB code was used to verify our Grid CANN implementation.

For HD and Grid recovery, we compute population RMSE over the N neurons of each clean reference state $\mathbf{h}_q^\star$:

$$e_q = \frac{1}{r} \sqrt{\frac{1}{N} \sum_{i=1}^{N} (h_{qi}^{\mathrm{out}} - h_{qi}^\star)^2}, \qquad r = \frac{1}{Q} \sum_{q=1}^{Q} \sqrt{\frac{1}{N} \sum_{i=1}^{N} (h_{qi}^\star)^2}.$$

The reference RMS r is computed within groups of $Q = 2000$ HD or 500 Grid states. Tables 1 and 3 pool the individual errors e_q from all groups and report their mean and SD across 10,000 HD or 2,500 Grid states per condition. Gaussian perturbations are $\mathbf{h}^{\mathrm{in}} = \mathbf{h}^\star + \sigma r \boldsymbol{\epsilon}$, with independent standard-normal entries in $\boldsymbol{\epsilon}$.

Table 3: **Paired recovery under Gaussian noise and quadrant bias.** Normalized population RMSE ↓, mean ± SD across test states. Bold marks the lower mean in each row.

Population and corruption	Amplitude	CAML	Decode and re-encode
HD, Gaussian σ	0.1	$\mathbf{0.014 \pm 0.009}$	0.020 ± 0.015
	0.5	$\mathbf{0.051 \pm 0.038}$	0.102 ± 0.077
	1.0	$\mathbf{0.102 \pm 0.078}$	0.203 ± 0.152
	2.0	$\mathbf{0.234 \pm 0.224}$	0.400 ± 0.291
Grid, Gaussian σ	0.1	0.078 ± 0.032	$\mathbf{0.007 \pm 0.004}$
	0.5	0.079 ± 0.032	$\mathbf{0.034 \pm 0.019}$
	1.0	0.081 ± 0.033	$\mathbf{0.069 \pm 0.038}$
	2.0	$\mathbf{0.088 \pm 0.037}$	0.137 ± 0.077
Grid, quadrant bias s	0.1	0.078 ± 0.032	$\mathbf{0.072 \pm 0.027}$
	0.5	$\mathbf{0.086 \pm 0.037}$	0.357 ± 0.128
	1.0	$\mathbf{0.109 \pm 0.053}$	0.686 ± 0.225
	2.0	$\mathbf{0.404 \pm 0.337}$	1.144 ± 0.296

CAML and learned baselines. CAML, DMHN-derived, and MHN use identical clean and corrupted states. HD uses Eq. (30) with $N = 64, \kappa = 4$, and Grid uses Eq. (32) with $N = 64^2, \kappa = 2$. CAML uses $M = 4096$ (HD) or 16384 (Grid), and RK4 with $dt = 0.05$ s for 25 s (HD) or 200 s (Grid). DMHN-derived uses symmetric weights $\mathbf{W} = \mathbf{U}\mathbf{U}^\top$ with $\mathbf{U} \in \mathbb{R}^{N \times 16384}$, no cue, leak, or bias, and update size 0.5. It runs for at most 200 updates, stopping when the maximum absolute state change is below 5×10^{-7}. MHN uses 16384 learned patterns and inverse temperature 8, repeatedly applying softmax retrieval for at most 200 updates until the maximum state change is below 10^{-6}.

CANN reference states. The HD and Grid CANNs use fixed recurrent kernels with $N = 64$ and $N = 64^2$, respectively. We generate clean references using their preparatory dynamics. CANNs share the standard-normal noise draws with the other methods, scaled by their own reference RMS.

Higher noise. Using the protocol of Table 1, HD errors at $\sigma = 1$ and 2 are 0.102 and 0.234 for CAML, compared with 0.230 and 0.409 for CANN. For Grid, CAML errors are 0.081 and 0.088, compared with 0.166 and 0.204 for CANN.

Restoration trajectories. Figure 3 uses 2000 perturbed states with absolute Gaussian noise SD 0.3 (HD) or 0.5 (Grid), and 256 clean controls. RK4 uses $dt = 0.05$ s for 25 s (HD) or 200 s (Grid). Embeddings are fixed linear readouts of raw activity.

E.2 Direct Decoding Comparison

Direct decoding maps each corrupted state to a coordinate and re-encodes it with the supplied tuning curves, without learned restoration. Table 3 uses the same states and recovery durations as above. For Grid quadrant bias, an activation of sr is added to the 961 neurons with preferred phases $\mu_{ix} > 0, \mu_{iy} > 0$.

Population-vector averaging attenuates independent Gaussian noise, making direct decoding particularly effective for the larger Grid population. Quadrant bias instead produces a systematic phase shift. CAML has lower error for quadrant bias at $s \geq 0.5$ and for every tested HD Gaussian noise level. For Grid Gaussian noise, CAML also has lower error at $\sigma = 2.0$.

Place recovery by energy.

Energy	Recovery MSE ($\times 10^{-3}$)
Fourier	1.74 ± 1.74
Quadratic	13566.00 ± 9664.13
Combined	$\mathbf{1.60 \pm 1.62}$

F Bump-Shifting Training and Evaluation

F.1 HD and Grid

For each periodic coordinate, finite differences of clean codes give target tangent fields:

$$\begin{aligned} \mathbf{g}_j^\star(\mathbf{x}) &= \frac{\mathrm{enc}(\mathbf{x}+\delta\mathbf{e}_j) - \mathrm{enc}(\mathbf{x}-\delta\mathbf{e}_j)}{2\delta}, \\ \mathcal{L}_{\mathrm{shift}} &= \sum_{j=1}^{d} \left[\frac{1}{N} \mathbb{E}_{\mathbf{x}} \|\mathbf{G}_{\boldsymbol{\phi},j}\, \mathrm{enc}(\mathbf{x}) - \mathbf{g}_j^\star(\mathbf{x})\|_2^2 + \frac{\lambda_1}{N^2} \|\mathbf{G}_{\boldsymbol{\phi},j}\|_{\mathrm{entry},1} \right]. \end{aligned} \tag{41}$$

Here $d = 1$ (HD) or 2 (Grid), each $\mathbf{G}_{\boldsymbol{\phi},j}$ is an $N \times N$ matrix, and $\lambda_1 = 0.001$. The finite-difference step is one quarter of the preferred-phase spacing. With the energy fixed, matrices are initialized by Xavier-uniform draws with gain 0.01 and trained for 2000 updates using batch size 4096 (HD) or 1024 (Grid), and AdamW. The learning rate decays from 10^{-3} to 10^{-5} by a cosine schedule, with zero weight decay, gradient-norm threshold 5, and betas $(0.9, 0.999)$ (HD) or $(0.9, 0.95)$ (Grid).

Table 2 uses 100 paired trajectories per velocity limit. Both dynamics and velocity inputs update at $dt = 0.05$ s. Trajectories start from clean zero-phase codes. Errors are time-averaged absolute wrapped phase differences over 2–20 s (HD) or 4–40 s (Grid). Grid errors are averaged across the two axes within each trajectory before computing the across-trajectory mean and SD. For each phase axis a, velocity starts at zero and follows

$$\omega_{a,k+1} = \mathrm{clip}\Big(\omega_{a,k} + 0.3\sqrt{dt}\,\epsilon_{a,k+1}, -v_{\max}, v_{\max}\Big), \qquad \epsilon_{a,k} \sim \mathcal{N}(0,1),$$

with $v_{\max} = 0.5$ or 1.0 rad/s. Grid uses independent innovations with the same distribution and bounds on both axes. Each velocity is held fixed during an RK4 step. Fixed and learned shifting receive identical inputs. Figure 4a,b shows 20 s fixed-shifting examples with velocity limit 0.5 rad/s. Panel f shows Grid tracking at the 1.0 rad/s per-axis limit.

The firing maps in Fig. 4c,d pool fixed-shifting walks with a 10 s burn-in. HD uses 32 walks of 250 s and 72 heading bins. Grid uses 256 walks of 100 s in a 5 m-radius arena, displaying the central 4 m region. Grid rates use 50×50 occupancy-normalized bins with Gaussian smoothing of SD 0.2 m.

F.2 Place

The population has 256 Gaussian tuning curves in a $1\,\mathrm{m} \times 1\,\mathrm{m}$ arena:

$$\mathrm{enc}_{\mathrm{place},i}(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}-\mathbf{c}_i\|_2^2}{2\ell^2}\right), \qquad \ell = 0.08\,\mathrm{m}. \tag{42}$$

The free region is defined by $x, y \in [0.1, 0.9]\,\mathrm{m}$, and the energy uses 8192 Fourier features. Field centres $\mathbf{c}_i$ are selected by farthest-point sampling on the free-map grid. Clean positions are drawn from a uniform free-map grid with independent Gaussian jitter of SD 0.02 m per coordinate. Proposals outside the free region are rejected and resampled.

For energy training and evaluation, the decoder accepts a log-space least-squares fit to the Gaussian tuning curves when at least four cells have $h_i > 0.01$, the RMS residual in log activity is below 0.5, and the fitted position is in the free region. Otherwise it returns the nearest encoded point on a free-map grid with spacing approximately 0.0204 m:

$$\mathrm{dec}_{\mathrm{place}}(\mathbf{h}) = \operatorname*{arg\,min}_{\mathbf{x} \in \mathcal{G}_{\mathrm{free}}} \|\mathbf{h} - \mathrm{enc}_{\mathrm{place}}(\mathbf{x})\|_2^2. \tag{43}$$

Energy training follows Appendix C. Place dynamics use $\dot{\mathbf{h}} = -\gamma \nabla E_{\boldsymbol{\theta}}(\mathbf{h}) + \mathbf{B}(\mathbf{h}, \mathbf{u})$, with $\gamma = 1$ for energy training and $\gamma = 2$ for shifting training and evaluation, using $dt = 0.05$ s. Recovery uses

absolute Gaussian noise SD 0.3 and 50 s integration. We report the mean and SD of the per-state MSE, $m_q = N^{-1}\|\mathbf{h}_q^{\mathrm{out}} - \mathbf{h}_q^{\star}\|_2^2$, across 20,000 test states.

Fixed shifting uses $\mathbf{B}_{\mathrm{fixed}} = -v_x\mathbf{D}_x\mathbf{h} - v_y\mathbf{D}_y\mathbf{h}$. The central-difference operators use the other field centres nearest to $\mathbf{c}_i \pm \delta\mathbf{e}_j$ along each spatial axis j, with $\delta = 0.05\,\mathrm{m}$. Learned shifting uses a shared MLP with hidden widths 1024, 512, 256 and GELU activations, followed by two 256-dimensional heads. The MLP takes activity as input and contributes $v_x\mathbf{G}_{\boldsymbol{\phi},x}(\mathbf{h}) + v_y\mathbf{G}_{\boldsymbol{\phi},y}(\mathbf{h})$ to the dynamics. Head outputs are bounded at 1.5 times the maximum finite-difference reference magnitude, calibrated using 2048 clean states and standard-normal velocities. With the restoring energy fixed, encoded trajectories supply targets $\mathbf{h}_k^{\star} = \mathrm{enc}_{\mathrm{place}}(\mathbf{x}_k)$ for the rollout loss

$$\mathcal{L}_{\mathrm{rollout}} = \frac{1}{K}\sum_{k=1}^{K}\frac{\|\mathbf{h}_k - \mathbf{h}_k^{\star}\|_2^2}{N}. \tag{44}$$

Training starts from random initialization and uses 5000 updates, batch size 4096, microbatch size 1024, and a curriculum increasing K from 1 to 20. AdamW uses learning rate 10^{-3}, weight decay 10^{-6}, gradient-norm threshold 10, and cosine decay to 10^{-5}. Training velocities have independent components drawn from $\mathcal{U}(-0.5, 0.5)\,\mathrm{m/s}$, with target positions restricted to the free square after each step. The checkpoint with lowest rollout MSE on 2048 fixed 20-step validation trajectories is selected.

Tracking uses 100 paired 20 s random walks, with target-wall reflections and velocities updated every 0.05 s. Both shifting methods use identical inputs and the same decoder. Position error is the full-trajectory mean Euclidean distance to the target.

Energy ablation. We compare Fourier energy, quadratic energy $E_0(\mathbf{h}) = -\frac{1}{2}\mathbf{h}^{\top}\mathbf{W}\mathbf{h}$ with symmetric $\mathbf{W}$, and their sum. The combined energy, used for the main Place results, has the lowest mean recovery error. The quadratic-only model exhibits activity growth despite decreasing energy, since the quadratic energy is not bounded below.